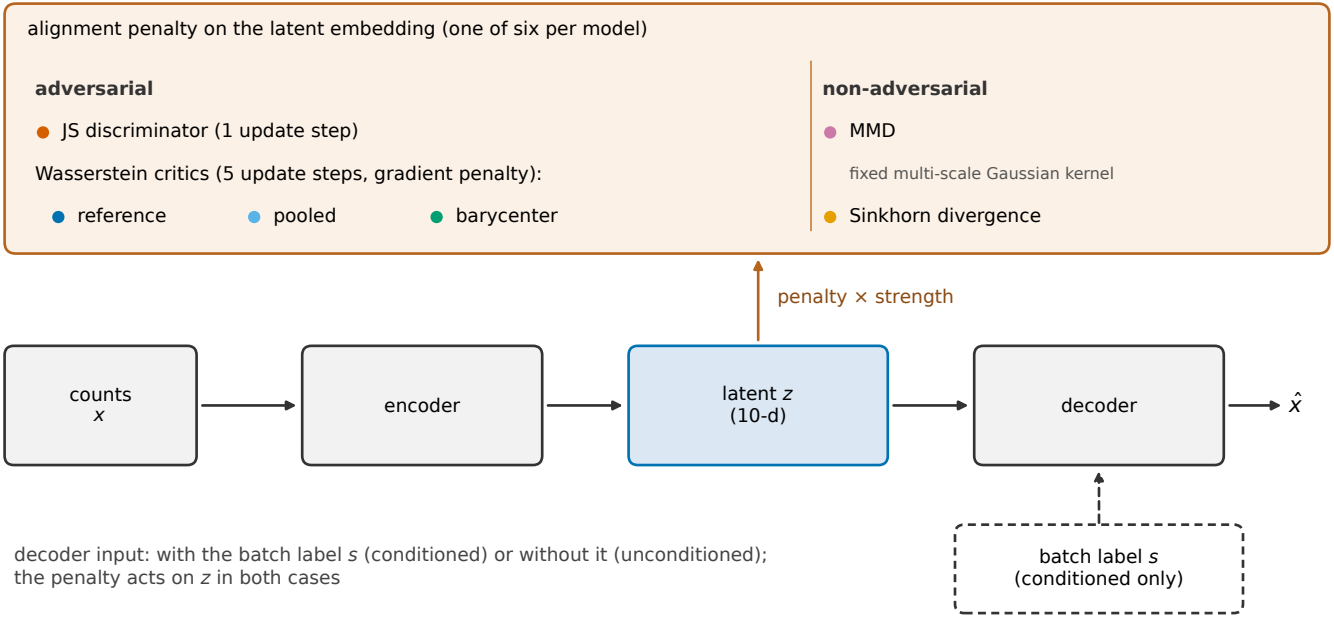

alignment penalty on the latent embedding (one of six per model)
adversarial
JS discriminator (1 update step)
Wasserstein critics (5 update steps, gradient penalty):
reference
pooled
barycenter
non-adversarial
MMD
fixed multi-scale Gaussian kernel
Sinkhorn divergence
penalty × strength
counts
x
encoder
latent z
(10-d)
decoder
x̂
batch label s
(conditioned only)
decoder input: with the batch label s (conditioned) or without it (unconditioned);
the penalty acts on z in both cases
loss = scVI loss + strength × penalty (strength 0 = plain scVI)